King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Final Exam Term 252

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. $\int_0^{\sqrt{\ln 16}} \int_x^{\sqrt{\ln 16}} \int_1^3 e^{y^2}\, dz\, dy\, dx =$

(a) $\sqrt{\ln 16} - 1$

(b) 15

(c) $\sqrt{\ln 16} + 1$

(d) $1 + 4 \ln 2$

(e) 16

2. The surface of a mountain is modeled by the equation

$$h(x, y) = 5000 - 0.001x^2 - 0.004y^2.$$

If a mountain climber is at the point $(500, 200, 4590)$, then which of the following vectors points in the direction of steepest descent (descend at the greatest rate)?

(a) $5\,\mathbf{i} + 8\,\mathbf{j}$

(b) $-5\,\mathbf{i} - 3\,\mathbf{j}$

(c) $-\mathbf{i} - 3\,\mathbf{j}$

(d) $-\mathbf{i} + 8\,\mathbf{j}$

(e) $5\,\mathbf{i} - 8\,\mathbf{j}$

3. The graph of the function

$$f(x, y) = \tfrac{1}{3}x^3 - x^2 + (x - 1)y^2$$

has

(a) two relative maxima and two relative minima

(b) one saddle point, two relative maxima and one relative minimum

(c) two saddle points and two relative maxima

(d) one saddle point, one relative maximum and two relative minima

(e) two saddle points, one relative maximum and one relative minimum

4. If $D = \{(x, y, z) : 0 \le z \le \sqrt{9 - x^2 - y^2},\ 0 \le y \le \sqrt{9 - x^2},\ 0 \le x \le 3\}$, then

$$\iiint_D (x^2 + y^2 + z^2)\, dz\, dy\, dx =$$

(a) $\int_0^{\pi/2} \int_0^{\pi/2} \int_0^3 \rho^4 \sin\phi \, d\rho \, d\phi \, d\theta$

(b) $\int_0^{\pi} \int_0^{\pi/2} \int_0^3 \rho^4 \sin\phi \, d\rho \, d\phi \, d\theta$

(c) $\int_0^{\pi/2} \int_0^{\pi/2} \int_0^3 \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$

(d) $\int_0^{\pi} \int_0^{\pi} \int_0^3 \rho^4 \sin\phi \, d\rho \, d\phi \, d\theta$

(e) $\int_0^{\pi/2} \int_0^{\pi} \int_0^3 \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$

5. The volume of the solid lying in the first octant and bounded by the graphs of $z = 8(1 - xy)$, $y = x$, $y = 1$ is equal to

(a) 0

(b) $\frac{1}{8}$

(c) 8

(d) 3

(e) $\frac{1}{3}$

6. Let **P** be the tangent plane to the surface $xyz = 10$ at the point $(1, 2, 5)$. Which of the following points is not on **P**?

(a) $(0, 0, 15)$

(b) $(0, 6, 0)$

(c) $(0, 4, 5)$

(d) $(1, 1, 1)$

(e) $(3, 0, 0)$

7. If $w = x\cos(yz)$, $x = s^2$, $y = t^2$, $z = s - 2t$, then the value of $\dfrac{\partial w}{\partial t}$ when $s = 4$ and $t = 1$ is equal to

(a) $-32\sin(2)$

(b) $16\sin(1)$

(c) 16

(d) -32

(e) 0

8. The distance between the two planes

$$\mathcal{P}_1: \ 2x + \sqrt{5}\, y + 4z = \sqrt{5} \qquad \mathcal{P}_2: \ 2x + \sqrt{5}\, y + 4z = 31\sqrt{5}$$

is equal to

(a) 4

(b) 0

(c) $6\sqrt{5}$

(d) $\sqrt{5}$

(e) $\dfrac{2}{\sqrt{5}}$

9. The minimum distance from the point $(-2, -3, 0)$ to the surface

$$z = \sqrt{3 - 2x - 2y}$$

is equal to

(a) $\sqrt{11}$

(b) $\sqrt{13}$

(c) 9

(d) 5

(e) 3

10. If $R = \{(x, y) : x^2 + y^2 \leq 9, \ x \geq 0, \ y \geq 0\}$, then $\iint_R (x + y)\, dA =$

(a) 27

(b) 54

(c) 18

(d) 15

(e) 9

11. If n is the number of intersection points between the two polar curves

$$r = 1 - 2\cos\theta \quad \text{and} \quad r = d \quad \text{where } 0 \leq \theta \leq 2\pi,$$

then which one of the following statements is FALSE?

(a) if $d = 3$, then $n = 1$

(b) if $d = 2.9$, then $n = 2$

(c) if $d = 1$, then $n = 3$

(d) if $d = 0.4$, then $n = 4$

(e) if $d = 3.5$, then $n = 1$

12. The volume of the solid bounded above by $z = 12 - x^2 - y^2$ and below by $z = 2x^2 + 2y^2$ is equal to

(a) 12π

(b) 36π

(c) 2π

(d) 6π

(e) 24π

13. Which one of the following points lies on the line that passes through the point $(-4, 5, 2)$ and is perpendicular to the plane given by $-x + 2y + z = 5$?

(a) $(-5, 7, 7)$

(b) $(0, 7, 3)$

(c) $(-3, 3, 3)$

(d) $(-3, 3, 1)$

(e) $(4, 7, 3)$

14. Let M and m denote, respectively, the maximum and minimum values of

$$f(x, y) = 2x^2 + 2y^2 - 4x - 4y$$

over the region R. The region R is the triangular region in the xy-plane with vertices $(0, 0)$, $(2, 0)$, and $(2, 4)$, and it includes its boundary. The value of $M + m =$

(a) 16

(b) 14

(c) -2

(d) 12

(e) -4

15. The parametric curve $x = 3\sec t,\ y = 4\tan t$ is represented in rectangular form by the equation

(a) $\dfrac{x^2}{9} - \dfrac{y^2}{16} = 1$

(b) $y = x^2 - 144$

(c) $\dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$

(d) $\dfrac{x}{4} + \dfrac{y}{3} = 1$

(e) $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$

16. Let $f(x,y) = 2x^3 - x^2y + y^2$, $P(-1,1)$ and $Q(2,-3)$. Then, the directional derivative of f at P in the direction of $\overrightarrow{PQ}$ is equal to

(a) 25

(b) 4

(c) 20

(d) -2

(e) 3

17. The minimum value of $f(x,y,z) = xyz + 228$ on the sphere $x^2 + y^2 + z^2 = 27$ is equal to

(a) 101

(b) 208

(c) 201

(d) 202

(e) 102

18. Let $f(x,y) = y - \sin x$. Which of the following vectors is normal to the level curve $f(x,y) = 2$ at the point $\left(\frac{\pi}{2}, 3\right)$?

(a) $\langle 1, -\frac{\pi}{2} \rangle$

(b) $\langle \frac{\pi}{2}, -3 \rangle$

(c) $\langle -1, 0 \rangle$

(d) $\langle \frac{\pi}{2}, 3 \rangle$

(e) $\langle 0, 1 \rangle$

19. $\int_1^e \int_0^y \frac{4}{x^2+y^2}\, dx\, dy =$

(a) $e \ln \pi$

(b) $e \ln 4$

(c) π

(d) $e^2\pi$

(e) $\pi \ln 4$

20. Let $f(x,y) = ax^2 + ay^2$, where $a > 0$ and $R = \{(x,y) : 0 \le x \le a,\ 0 \le y \le a\}$. If the average value of f over R is 18, then $a =$

(a) 6

(b) 9

(c) 3

(d) 2

(e) 18

21. $\int_{-1}^{0} \int_{4x}^{-4x^2} f(x,y)\,dy\,dx =$

(a) $\int_{-4}^{0} \int_{-\sqrt{-y/4}}^{-y/4} f(x,y)\,dx\,dy$

(b) $\int_{-4}^{0} \int_{-\sqrt{-y/4}}^{y/4} f(x,y)\,dx\,dy$

(c) $\int_{-4}^{0} \int_{-\sqrt{y/4}}^{y/4} f(x,y)\,dx\,dy$

(d) $\int_{-4}^{0} \int_{\sqrt{-y/4}}^{y/4} f(x,y)\,dx\,dy$

(e) $\int_{-4}^{0} \int_{\sqrt{y/4}}^{-y/4} f(x,y)\,dx\,dy$

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	Integrate z, swap order: $\int_0^{\sqrt{\ln 16}} 2ye^{y^2}\,dy = 16 - 1 = 15$.
2	**(a)**	$-\nabla h = \langle 1, 1.6\rangle \parallel 5\mathbf{i} + 8\mathbf{j}$.
3	**(e)**	Critical points $(0,0)$ max, $(2,0)$ min, $(1,\pm 1)$ saddles.
4	**(a)**	First octant of the ball $\rho \le 3$; integrand $\rho^2 \cdot \rho^2 \sin\phi = \rho^4 \sin\phi$.
5	**(d)**	$V = \int_0^1 \int_0^y 8(1 - xy)\,dx\,dy = 8(\frac{1}{2} - \frac{1}{8}) = 3$.
6	**(d)**	Plane $10x + 5y + 2z = 30$; $(1,1,1)$ gives 17, so it is not on it.
7	**(a)**	$w_t = w_y \cdot 2t + w_z \cdot (-2) = -64\sin 2 + 32\sin 2 = -32\sin 2$.
8	**(c)**	$D = \frac{\lvert\sqrt{5} - 31\sqrt{5}\rvert}{\sqrt{25}} = 6\sqrt{5}$.
9	**(a)**	$d^2 = (x+2)^2 + (y+3)^2 + 3 - 2x - 2y$ is minimized at $(-1,-2)$: $d^2 = 11$.
10	**(c)**	Polar: $\int_0^3 r^2 dr \cdot \int_0^{\pi/2} (\cos\theta + \sin\theta) d\theta = 9 \cdot 2 = 18$.
11	**(e)**	r ranges over $[-1, 3]$, so $d = 3.5$ gives $n = 0$, not 1.
12	**(e)**	Meet at $r = 2$: $V = 2\pi \int_0^2 (12 - 3r^2) r\,dr = 24\pi$.
13	**(d)**	Line: $(-4 - t,\ 5 + 2t,\ 2 + t)$; $t = -1$ gives $(-3, 3, 1)$.
14	**(d)**	Interior $(1,1)$: -4; vertex $(2,4)$: 16; edges give nothing lower than -4, so $M + m = 12$.
15	**(a)**	$\sec^2 t - \tan^2 t = 1$ gives $\frac{x^2}{9} - \frac{y^2}{16} = 1$.
16	**(b)**	$\nabla f(P) = \langle 8, 1\rangle$, $\mathbf{u} = \langle \frac{3}{5}, -\frac{4}{5}\rangle$: $D_{\mathbf{u}} f = 4$.
17	**(c)**	$x^2 = y^2 = z^2 = 9$, so $xyz \in \{\pm 27\}$; minimum $-27 + 228 = 201$.
18	**(e)**	$\nabla f = \langle -\cos x, 1\rangle = \langle 0, 1\rangle$ at $x = \frac{\pi}{2}$.
19	**(c)**	Inner integral $= \frac{\pi}{y}$, outer $= \pi \ln e = \pi$.
20	**(c)**	Average $= \frac{2a^3}{3} = 18 \Rightarrow a = 3$.
21	**(b)**	Left curve $x = -\sqrt{-y/4}$, right curve $x = y/4$, $-4 \le y \le 0$.